

Habiba Morsy

CONTACT

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Profiles: [Linkedin](#), [Github](#)

TEACHING EXPERIENCE

Graduate Teaching Assistant

University of Virginia

- Courses:
 - Machine Learning in Systems and Network Security: Master's Level 06/2026 – 08/2026
 - Introduction to Deep Learning: Undergraduate Level 01/2026 – 05/2026
 - Computational Probability: Undergraduate Level 09/2025 – 12/2025

Student Assistant

Kean University

09/2023 – 05/2024

- Tutored students in Python and Java programming concepts and debugging techniques.
- Simplified data structures, algorithms, and OOP principles for diverse learning needs.

WORK EXPERIENCE

Software Developer Intern

05/2024 – 08/2024

RCSB Protein Data Bank

- Developed and implemented a Python wrapper for RCSB PDB's data API to facilitate streamlined access and integration of 3D macromolecular structure data
- Contributed to the open-source community by publishing the Python package on GitHub and PyPI
- Tools and libraries used: GraphQL, rustworkx, networkx, Microsoft Azure, Github, PyPI
- [Github Repository](#), [PyPI Package](#)

RESEARCH EXPERIENCE

Independent Research

09/2023 – 05/2024

American Council on Education & Kean University

- Developed an application using LLMs and RAG to handle CCIHE queries, dynamically updating the database with new Q&A pairs
- Integrated LLMs into a React dashboard to enhance user experience and ensure secure interactions

AI4ALL Student

09/2022 – 06/2023

Kean University

- Explored AI fundamentals, including machine learning types, gradient descent, and neural networks.
- Investigated ethical implications of AI and data processing.
- Gained expertise in neural networks, covering artificial neurons, activation functions, CNNs, and RNNs.
- Built machine learning models with TensorFlow and PyTorch, showcasing hands-on AI development skills.

Undergraduate Research Assistant

01/2022 – 05/2023

Kean University, Advisor: Dr. David Joiner, Acting Associate Dean

- Developed custom Unity applications for interactive data exploration
 - Implemented complex data interactions and user interfaces in VR environments using Unity scripting
 - Created and optimized 3D models for VR using Blender and Maya
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EDUCATION

University of Virginia

08/ 2025 – 05/ 2030

Doctor of Philosophy – PhD in Data Science

Kean University

M.S Computational Science and Engineering

01/ 2025 – 06/2025

(one semester – transferred to UVA)

- Master's Thesis: Developing Large Language Models to Automate Quality Assurance of Digital Library Metadata (completed & published)

B.S. in Science and Technology

09/ 2021 – 12/ 2025

Applied Mathematics Track, Computer Science Minor

- Dean's List 2021-2025
 - Vice President, Association for Computing Machinery
 - Treasurer, Association for Computing Machinery – Women
 - Secretary, Arab Culture Club
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AWARDS

- Generation Google Scholar
 - National Cyber Scholarship Scholar
 - NJ Seal of Biliteracy, French, English, Arabic
 - Distinction in Research Award – Kean University
 - STEM Scholarship Recipient
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PUBLICATION

[RCSB Protein Data Bank: Python Software Package for Accessing RCSB PDB APIs](#)

- Journal of Molecular Biology

[PC-ED: Testing Automated Agents to Assess the Quality of Training Resource Metadata](#)

- The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC25)

[Towards Foundation Models for Shock Compression of Solid Reactive Materials](#)

- 2026 International Annual Conference of the Fraunhofer ICT
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POSTER PRESENTATIONS

- International Annual Conference of the Fraunhofer ICT – 2026
- Research Days – Kean University 2022 – 2024
- ABRCMS Conference 2024
- Rutgers RISE REU Symposium 2024